

MAE 3127

Fluid Mechanics Laboratory



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Topics covered today:

1. Tin Foil Boat Data Analysis using Python
2. Manometers



School of Engineering
& Applied Science

THE GEORGE WASHINGTON UNIVERSITY

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Photo: Kartik Bulusu

Points to ponder on after you leave the recitation today:

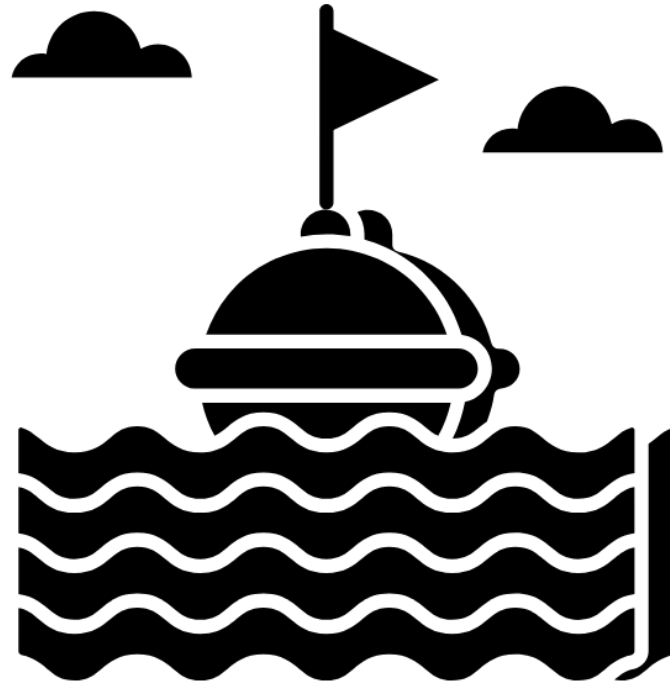
- Take all the notes you want, ask all the questions you want.
- You are encouraged to study in groups, but you may *not* show HWs among yourselves.
- Under *no circumstances* may you look at another student's HWs, or look for HW answers anywhere other than in the text and in the notes and handouts provided in class or links provided on the course website. If you do, it is considered an academic integrity violation!
- **Collaborate but don't violate!**

On deliverables:

- **Deadline for submitting P-set00 and P-set01** is 09/06.
- You need to **upload two PDFs, one for each P-set**, on Blackboard.
- **Please write clearly and legibly and upload clear images** to be graded.
- Office hours will be made available to you soon.
- Check into Ed discussion board: <https://edstem.org/us/courses/83318/discussion>



Tin Foil Boat Data Analysis using Python



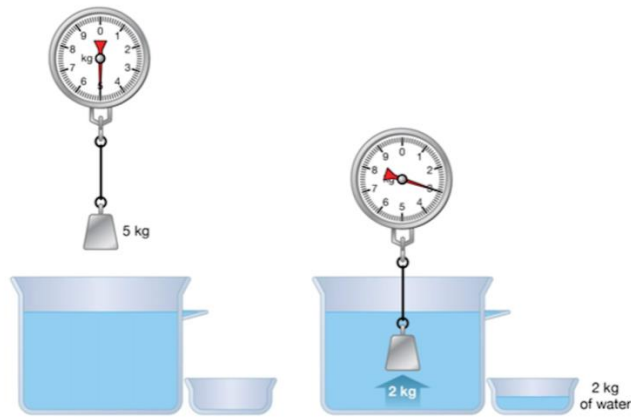
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Buoyancy principle:

An object floats when the upward buoyant force equals or exceeds the downward force from its weight.

Archimedes' principle

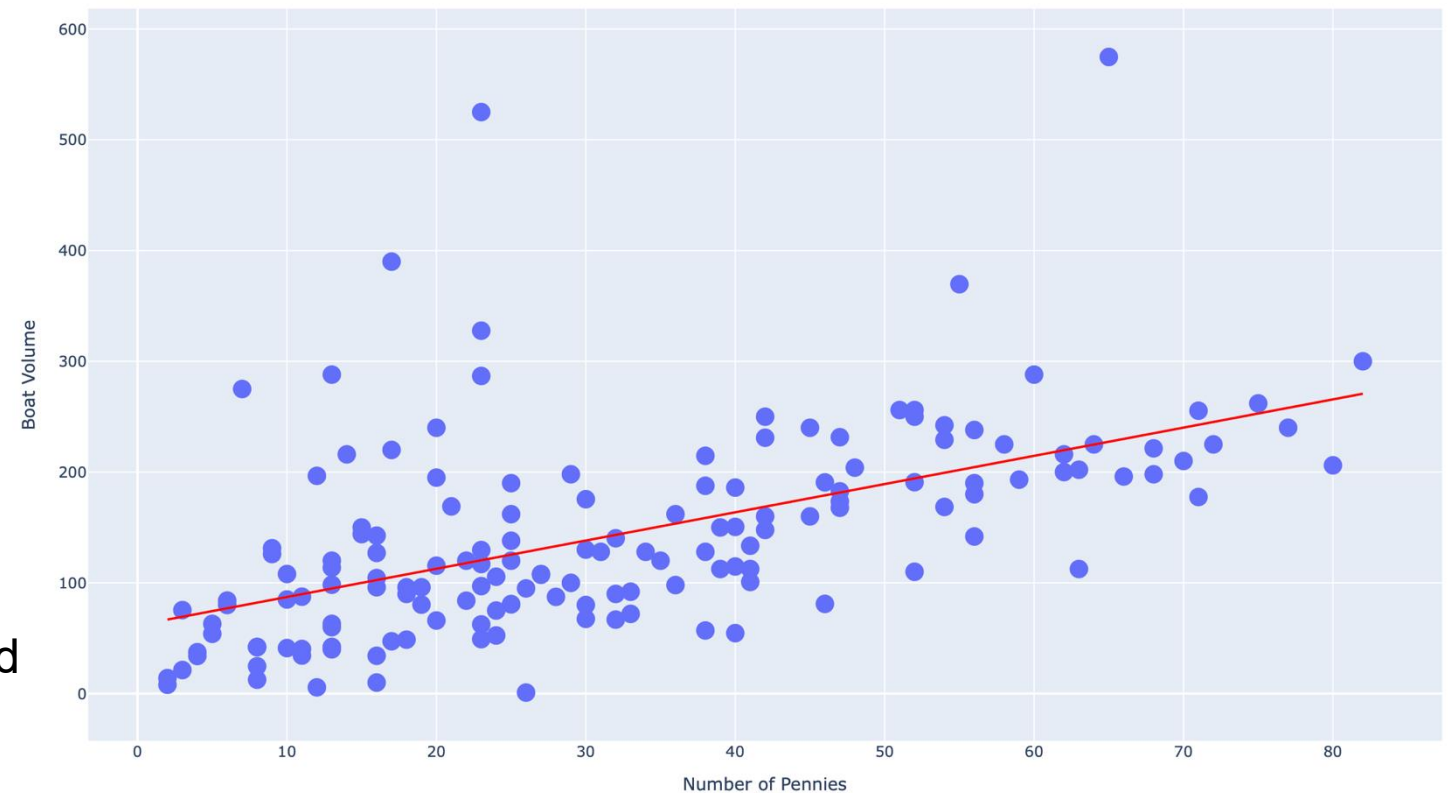


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The **buoyancy principle**—formulated as Archimedes' principle—states that the buoyant force acting on an object immersed in a fluid is equal to the weight of the fluid displaced by that object.

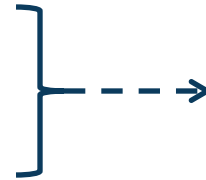
Mathematically we note.

$$\mathbf{F}_B = -\mathbf{F}_g = -\rho V \mathbf{g}$$

Number of Pennies Used vs. Boat Volume (cm³)

A simple Python program to plot data and calculate the correlation

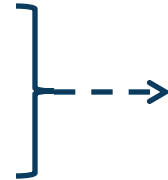
```
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import pearsonr
```



Import libraries that are relevant for loading data from CSV-file, plotting and calculating statistics

Note: You may have to install these libraries using the requirements.txt file provided to you

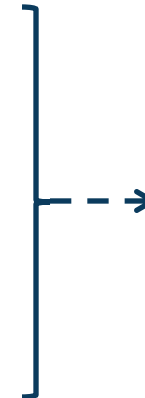
```
# Load data (include the file in the
# same folder as the program itself)
df = pd.read_csv('boat_data.csv')
```



Create a dataframe by loading the data from the CSV-file.

Note: The data file was provided to you

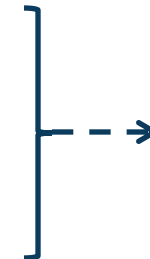
```
# Scatter plot
plt.figure(figsize=(8,6))
plt.scatter(df['Number of Pennies'], df['Boat Volume'], alpha=0.7)
plt.title('Number of Pennies vs Boat Volume')
plt.xlabel('Number of Pennies')
plt.ylabel('Boat Volume <cm^3>')
plt.grid(True)
plt.tight_layout()
plt.show()
```



Create a scatter plot from the data loaded into a dataframe.

Provide a title, x- and y-label, invoke a grid and show the plot

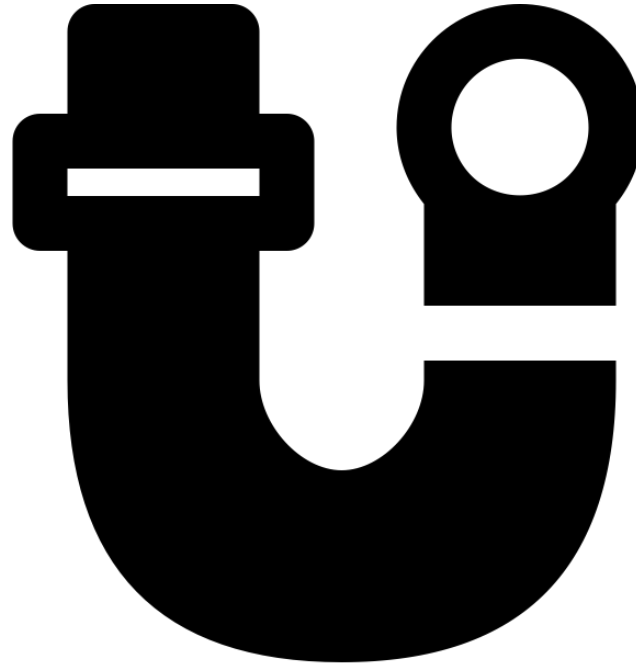
```
# Calculate Pearson correlation
correlation, p_value = pearsonr(df['Number of Pennies'], df['Boat Volume'])
print('Correlation Coefficient:', correlation)
print('P-value:', p_value)
```



Calculate the Pearson's correlation coefficient and the p-value of the data



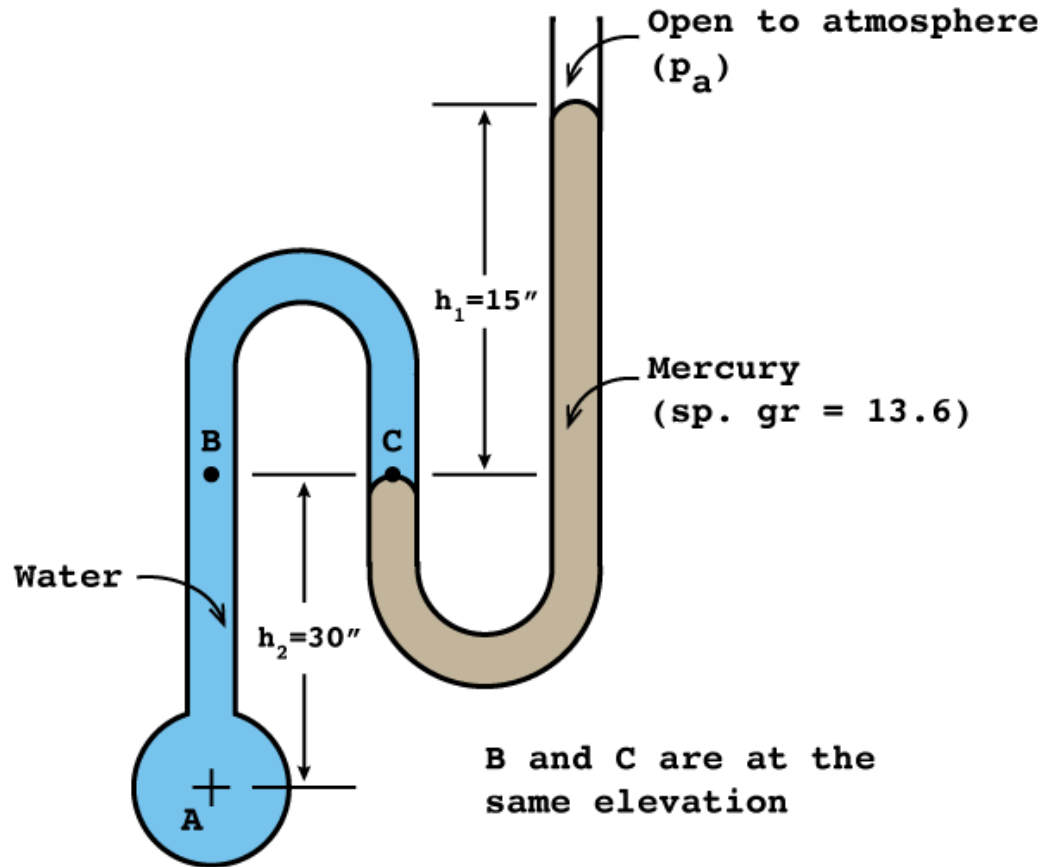
Manometers



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R0-P010: Determine the gage pressure [psig] at point A.



Step 1

Given: $(sp. gr)_{Hg} = 13.6$

We know: $\gamma_{Hg} = (sp. gr)_{Hg} \times (\gamma_{H_2O})$

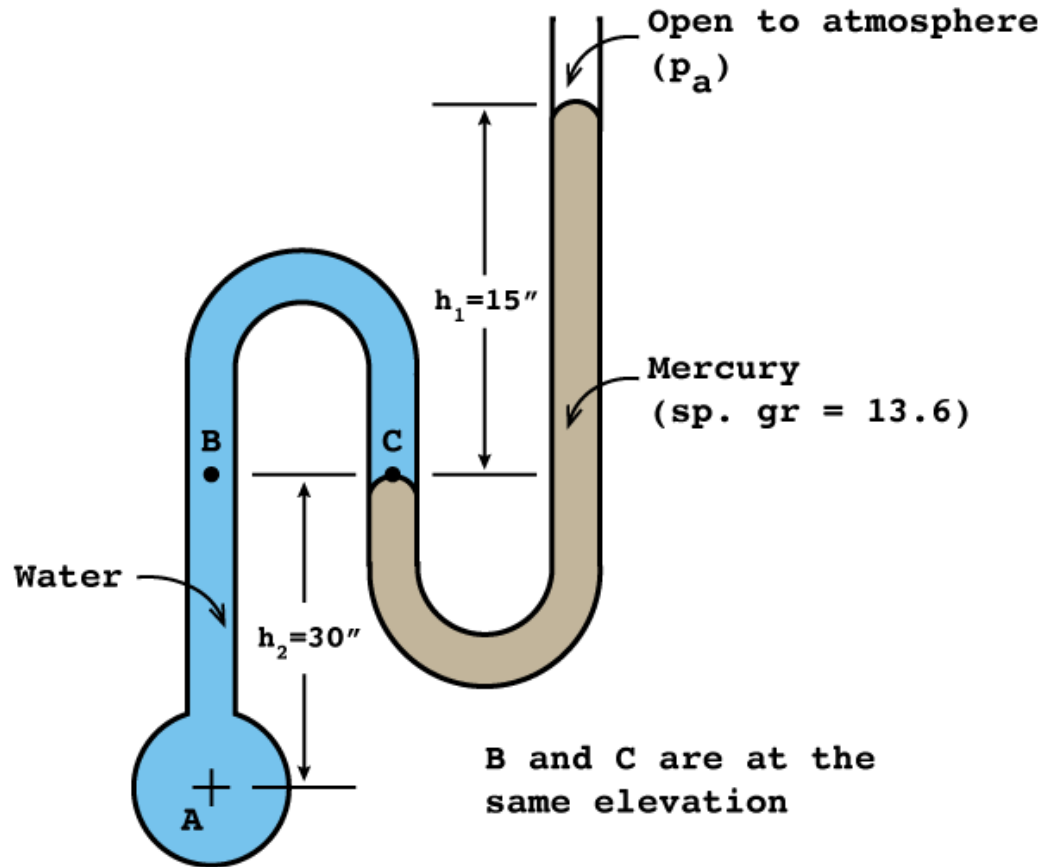
$$\gamma_{H_2O} = 62.4 \frac{lbf}{ft^3}$$

Calculate: γ_{Hg}



[Contd.]

R0-P010: Determine the gage pressure [psig] at point A.



Step 2: Strategy

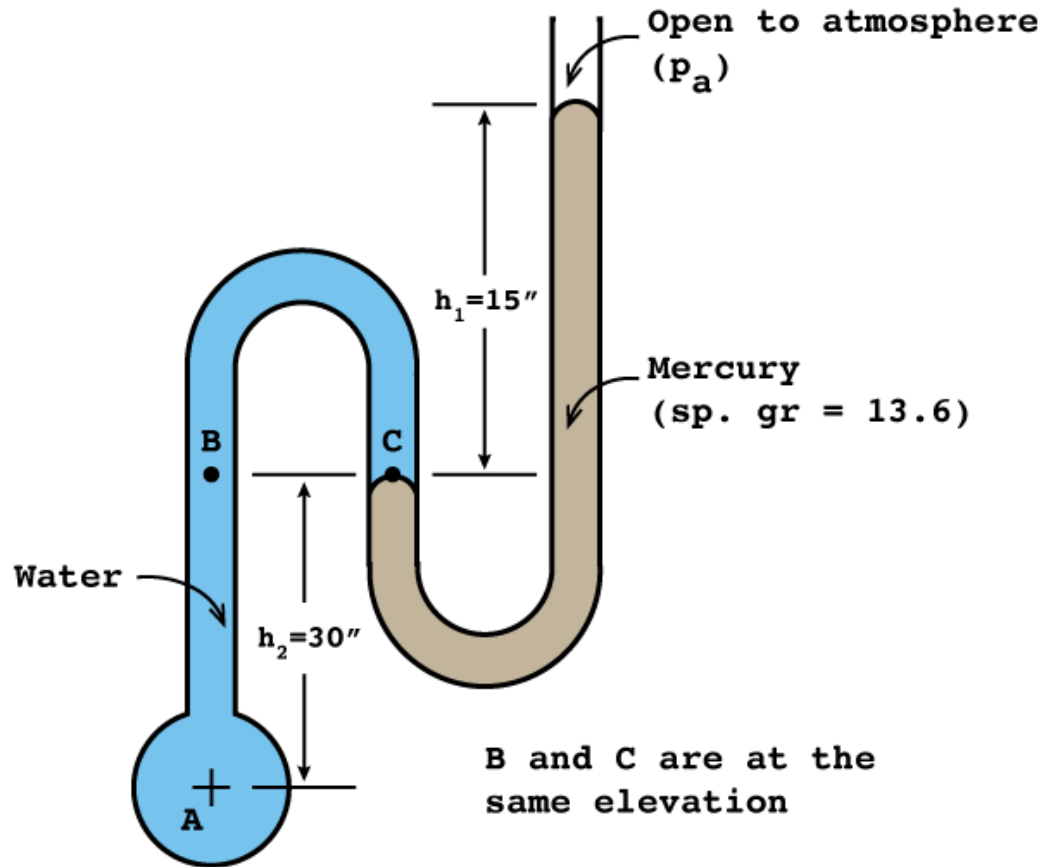
- Look for two points in the vertical segment
 - With equal pressures,
 - That must be at the same elevation, and
 - Joined by the same fluid.
- Compute the pressures at those points separately along each leg
 - p_B and p_C in this problem
- Equate the pressures and solve
 - $p_B = p_C$



[Contd.]

R0-P010: Determine the gage pressure [psig] at point A.

$$P_B = P_C$$



$$P_B = P_A - \gamma_{H_2O} h_2$$

$$P_C = P_{atm} + \gamma_{Hg} h_1$$

$$P_A - \gamma_{H_2O} h_2 = P_{atm} + \gamma_{Hg} h_1$$

$$P_A = \cancel{P_{atm}} + \gamma_{H_2O} h_2 + \gamma_{Hg} h_1$$
$$= \left(62.4 \frac{\text{lb}_f}{\text{ft}^3} \right) \left(30 \text{ in} \times \frac{\text{ft}}{12 \text{ in}} \right) + \left(13.6 \right) \left(62.4 \frac{\text{lb}_f}{\text{ft}^3} \right) \left(\frac{15 \text{ in} \times \text{ft}}{12 \text{ in}} \right)$$

$$P_A = 1216.8 \frac{\text{lb}_f}{\text{ft}^2} = 8.45 \text{ psig}$$

